Lecture 15

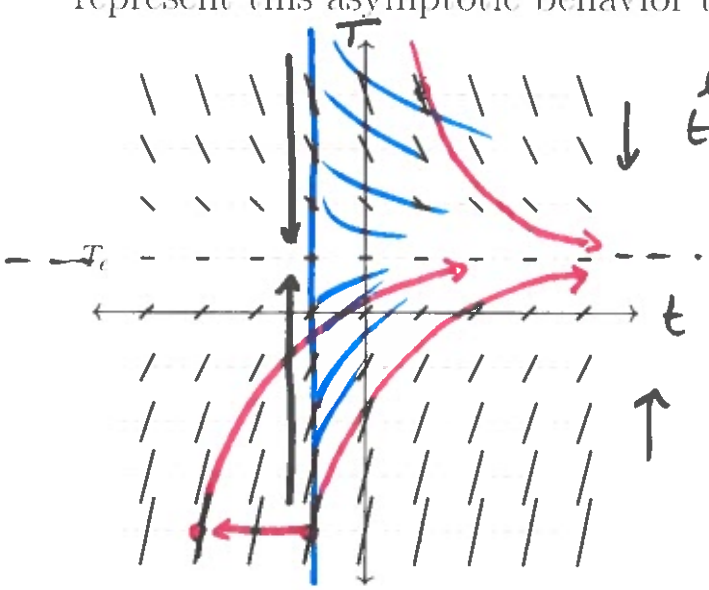
Phase Diagrams and the Phase Plane

We have been studying how to solve both first and second order equations. Of course, not every equation can be solved or has an explicit solution and we have employed numerical methods and the use of direction fields to help us understand these situations for first order equations. In this lecture, we will explore these ideas for second order equations to help us understand the qualitative information about our solutions without explicitly finding them.

To begin, we will consider first-order equations again. Suppose we are considering the equation

separable
$$\frac{dT}{dt} = k(T_e - T) \quad \Rightarrow \quad \overbrace{T(t) = T_e + (T_0 - T_e)e^{-kt}}^{\text{solution}}$$

given by Newton's Law of Cooling. We can note that $\lim_{t\to\infty} T(t) = T_e$. We can represent this asymptotic behavior through the use of a direction field.



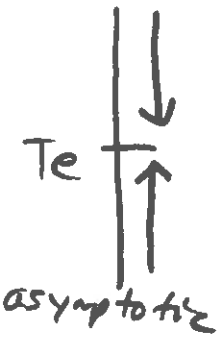
$$\lim_{t\to\infty} T(t) = \lim_{t\to\infty} T_e + (T_0 - T_e)\cancel{e^{-kt}}^{\to 0}$$
$$= T_e$$

$T > T_e, \quad T' < 0$

$T < T_e, \quad T' > 0$

However, we can note that the differential equation for Newton's Law of Cooling is **autonomous** (Recall: this means there are no instances of the independent variable, t, in the right hand side of the equation). So all horizontal shifts of the flows in the direction field (particular solutions) have similar asymptotic behavior (limits are the same). So in essence, only movement along the dependent variable axis affects the nature of the solution.

Therefore, we can condense our image of the direction field to a single axis and disregard the independent variable axis all together. This is called a **Phase Diagram** for the equation.



For

$$\frac{dT}{dt} = k(T_e - T) = 0$$

$$\Rightarrow T_e = T$$

we have the Phase Diagram:

$$\begin{array}{ccccc} T' > 0 & & T = T_e & & T' < 0 \\ \hline T < T_e & \longrightarrow & | & \longleftarrow & T > T_e \end{array}$$

equilibrium
stable

We see that $T = T_e$ when the right hand side of the differential equation $k(T_e - T)$ equals zero. Such values are called **critical points**. We usually refer to these constant solutions as **equilibriums**. The nature of any solution to Newton's Law of Cooling is to asympotically move towards the equilibrium $T = T_e$ at the critical point T_e.

Example: Consider the Logistic equation

$$\frac{dP}{dt} = P(\underset{\substack{\uparrow \\ K \text{ carrying capacity}}}{2} - P) \Rightarrow \overbrace{P(t) = \frac{2P_0}{P_0 + (2-P_0)e^{-2t}}}^{\text{solution}}$$

P_0 initial population

As $\lim_{t\to\infty} P(t) = \lim_{t\to\infty} \frac{2P_0}{P_0 + (2-P_0)\cancel{e^{-2t}}^{\to 0}} = \frac{2P_0}{P_0} = 2 \quad (P_0 > 0)$

$\frac{dP}{dt} = P(2-P)$ autonomous $\quad P(2-P) = 0$

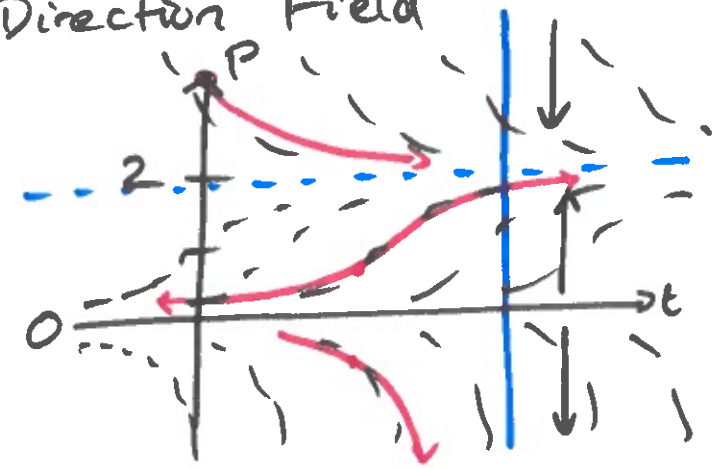
$\Rightarrow P = 0, \quad P = 2$

$\uparrow$ carrying capacity

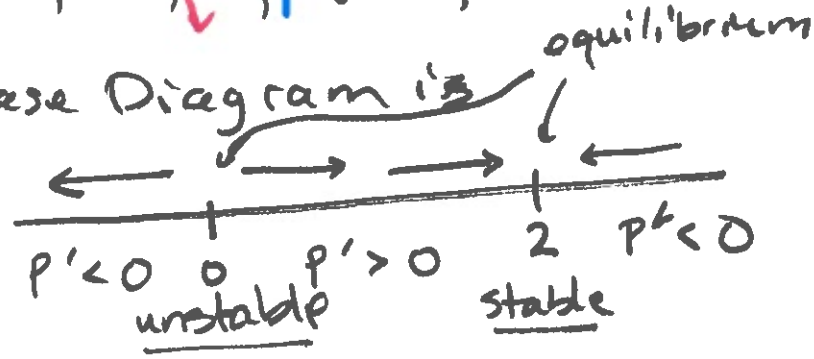
What if $P_0 \leq 0$, $P_0 = 0$, $P(t) = 0$

$P_0 < 0$, $P' < 0$

Direction Field

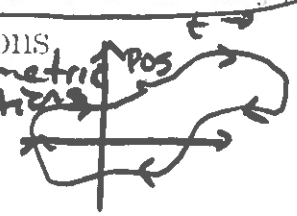


The Phase Diagram is

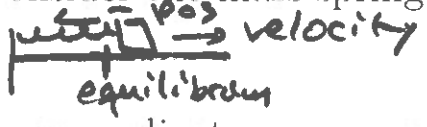


In the previous examples, the "movement" of the data (such as population, temperature, etc.) is along a single axis. If we wish to model movement using more than one variable (such as position and time, or position and velocity) we will need to expand our understanding to higher dimensions.

parametric equations



Consider the mass-spring oscillator homogeneous equation



$$my'' + by' + ky = 0$$

we can adjust our perspective by considering both the position, $y(t)$, and its velocity, $v(t) = y'(t)$. Using both y and v as variables we can rewrite our equation as

1st order in v, y

$$my'' + by' + ky = 0 \quad \Rightarrow \quad mv' + bv + ky = 0 \quad \Rightarrow \quad v' = -\frac{k}{m}y - \frac{b}{m}v$$

and combining this with $y' = v$ we have the system of two first-order equations

$$\begin{cases} y' = v \\ v' = -\dfrac{k}{m}y - \dfrac{b}{m}v \end{cases} = \begin{cases} \dfrac{dy}{dt} = v = f(y, v) \\ \dfrac{dv}{dt} = -\dfrac{k}{m}y - \dfrac{b}{m}v = g(y, v) \end{cases}$$

autonomous

which relates together the position and velocity through time. Note that both of these equations are autonomous as they do not involve the independent variable t.

In general we can consider a system of two first-order autonomous equations

$$\begin{cases} \dfrac{dx}{dt} = f(x, y) \\ \dfrac{dy}{dt} = g(x, y) \end{cases} \tag{15.1}$$

Just as with the single equation case, we wish to understand the nature of solutions (we usually write as pairs $(x(t), y(t))$) either through their explicit expressions or through their asymptotic behavior. parametric

As the equations in (15.1) are autonomous any solution $(x(t), y(t))$ is unchanged by a shift in the independent variable t. So, in other words, the relationship between the two functions x and y determine the nature of the solutions.

Definition: Given the first-order autonomous system (15.1) we can construct the equation

implizit

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g(x,y)}{f(x,y)}$$

$$\begin{cases} \frac{dx}{dt} = f(x,y) \\ \frac{dy}{dt} = g(x,y) \end{cases}$$

called the **phase-plane equation** of (15.1).

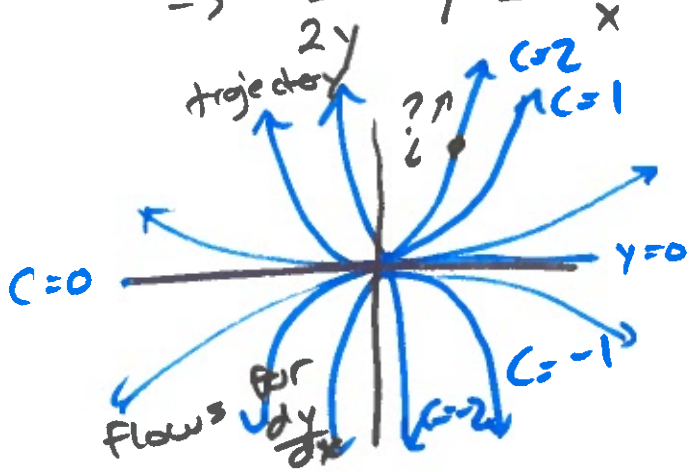
Example: Consider the system

$$\begin{cases} dx/dt = -x & = f(x,y) \\ dy/dt = -2y & = g(x,y) \end{cases}$$

Write the phase-plane equation and solve it as a separable equation.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2y}{-x} \quad \text{separable}$$

$$\Rightarrow \frac{1}{2y}\,dy = \frac{1}{x}\,dx \Rightarrow \int \frac{1}{2}\cdot\frac{1}{y}\,dy = \int \frac{1}{x}\,dx + C$$



$$\Rightarrow \frac{1}{2}\ln|y| = \ln|x| + C$$

$$\Rightarrow \ln|y| = 2\ln|x| + C$$

$$\Rightarrow \ln|y| = \ln|x^2| + C \quad \to e^{\ln(x^2)}\cdot e^{C}$$

$$\Rightarrow \boxed{y = Cx^2} \quad \to x^2\cdot C$$

We can treat the phase-plane equation as a basis for a direction field based upon the functions x and y (or thought of as vectors $\langle x'(t), y'(t)\rangle$, where direction is determined by $x'(t)$ and $y'(t)$.

A graph of the solution $(x(t), y(t))$ as a flow in the direction field is called a **trajectory**. The implicit solutions to the phase-plane equation give the trajectories of the system.

Definition: Consider the system

$$\begin{cases} \dfrac{dx}{dt} = f(x,y) = 0 \\ \\ \dfrac{dy}{dt} = g(x,y) = 0 \end{cases}$$

A point (x_0, y_0) where $f(x_0, y_0) = 0$ and $g(x_0, y_0) = 0$ is called a **critical point** (**equilibrium point**) of the system. The collection of all critical points is called the **critical point set**.

Note: If we have a critical point (x_0, y_0) then the equations $x(t) = x_0$, $y(t) = y_0$ form a solution to the system called an **equilibrium solution**. This is analogous to the equilibriums in the single equation case.

Example: We have seen that for the system

$$\begin{cases} dx/dt = -x \\ dy/dt = -2y \end{cases}$$

The phase plane equation and its solutions are

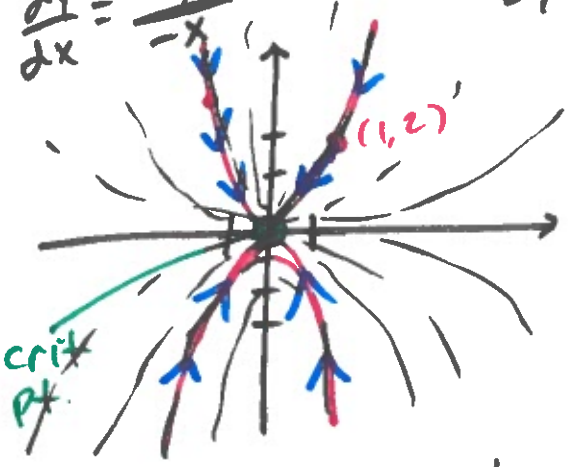
$$\frac{dy}{dx} = \frac{-2y}{-x} \qquad , y = Cx^2$$

Find any critical point(s) and sketch the direction field along with the trajectories at $(1, 2)$ and $(1, -2)$. What is the asymptotic behavior of the trajectories?

critical points $-x = 0 \Rightarrow x = 0$ point $(0,0)$ critical point.

$-2y = 0 \Rightarrow y = 0$

$\frac{dy}{dx} = \frac{-2y}{-x}$



$(1,2)$: $2 = C \cdot 1^2 \Rightarrow C = 2$

$x \quad y$

$y = 2x^2$

$1, 2, x > 0, y > 0 \Rightarrow \frac{dx}{dt} < 0$

$\frac{dy}{dt} < 0$

$x < 0, y > 0$

$\frac{dx}{dt} > 0, \frac{dy}{dt} < 0$

$t \to \infty$, move towards $(0,0)$ critical point

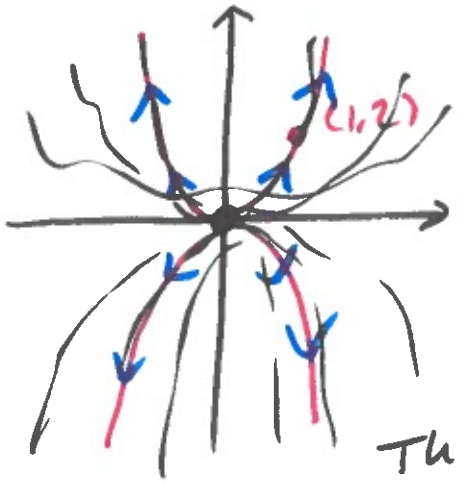
Example: Consider the system

$$\begin{cases} dx/dt = x \\ dy/dt = 2y \end{cases}$$

Phase-Plane Equation: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2y}{x}$

$\Rightarrow \frac{1}{2y} dy = \frac{1}{x} dx \Rightarrow y = Cx^2$

Same flows except the direction is reversed



In this case we move away from the critical point.

This is called a <u>Phase Plane</u>

Example: Consider the system

$$\begin{cases} \dfrac{dx}{dt} = 5x - 3y - 2 \\[2ex] \dfrac{dy}{dt} = 4x - 3y - 1 \end{cases}$$

To find the critical point(s) we solve $\begin{cases} 5x-3y-2=0 \\ 4x-3y-1=0 \end{cases}$

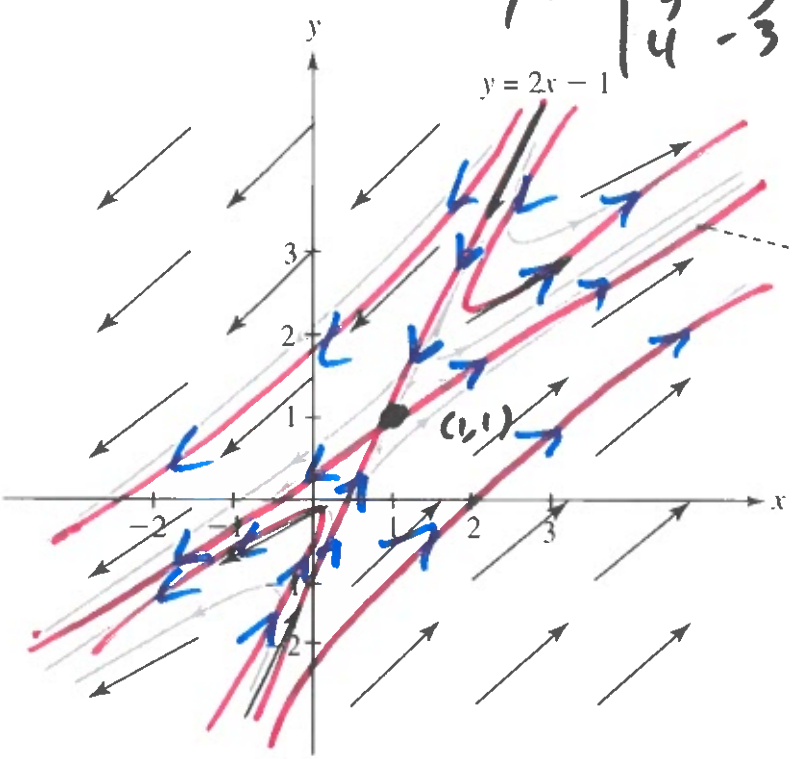
$\Rightarrow \begin{cases} 5x-3y=2 \\ 4x-3y=1 \end{cases}$ Cramer's Rule

$\begin{bmatrix} 5 & -3 \\ 4 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \Rightarrow$

$$x = \frac{\begin{vmatrix} 2 & -3 \\ 1 & -3 \end{vmatrix}}{\begin{vmatrix} 5 & -3 \\ 4 & -3 \end{vmatrix}} = \frac{2(-3)-1(-3)}{5(-3)-(4)(-3)} = \frac{-6+3}{-15+12} = \frac{-3}{-3} = 1$$

$$y = \frac{\begin{vmatrix} 5 & 2 \\ 4 & 1 \end{vmatrix}}{\begin{vmatrix} 5 & -3 \\ 4 & -3 \end{vmatrix}} = \frac{5(1)-4(2)}{5(-3)-4(-3)} = \frac{-3}{-3} = 1$$

crit. pt: $(1,1)$



movement both into and out of the crit. pt.

<u>saddle</u>

Theorem: Let $(x(t), y(t))$ be solutions on $[0, \infty)$ to the system

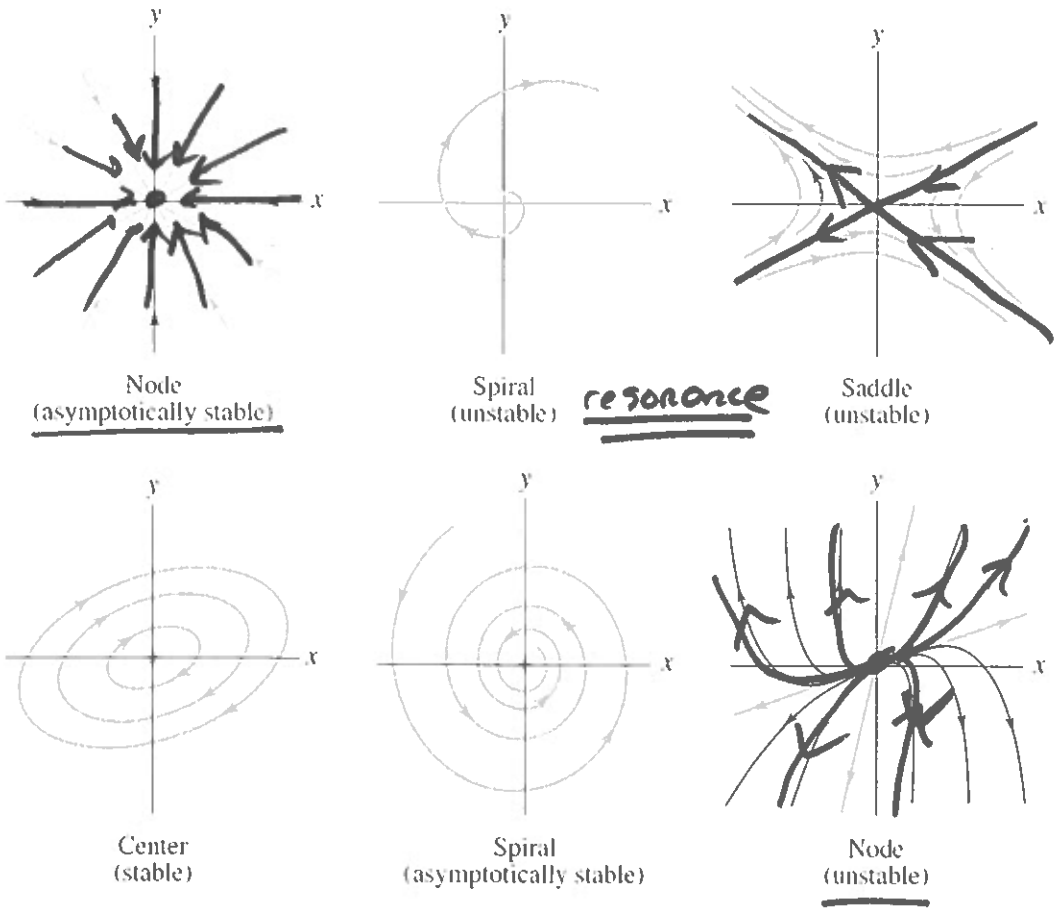
$$\begin{cases} dx/dt = f(x, y) \\ dy/dt = g(x, y) \end{cases}$$

where $f(x, y)$ and $g(x, y)$ are continuous. If

$$\lim_{t\to\infty} x(t) = x^\star, \quad \lim_{t\to\infty} y(t) = y^\star$$

exist and are finite, then $(x^\star, y^\star)$ is a critical point.

Below is a collection of graphs depicting the various types of critical points. We have so far seen asymptotically stable nodes, unstable nodes, and saddles.



Example: If we consider the homogeneous mass-spring oscillator with no damping.

$b=0$

$$my'' + ky = 0$$

we can write $v = y'$ and $mv' = -ky$ or $v' = -\frac{k}{m}y$ giving us the system

$$\begin{cases} \dfrac{dy}{dt} = v \\[2ex] \dfrac{dv}{dt} = -\dfrac{k}{m}y \end{cases}$$

Phase-Plane Equation: $\dfrac{dv}{dy} = \dfrac{-\frac{k}{m}y}{v}$ separable

$\Rightarrow v\,dv = -\frac{k}{m}y\,dy \Rightarrow \int v\,dv = \int -\frac{k}{m}y\,dy + C$

$\Rightarrow \frac{1}{2}v^2 = -\frac{k}{m}\cdot\frac{1}{2}y^2 + C$ implizit

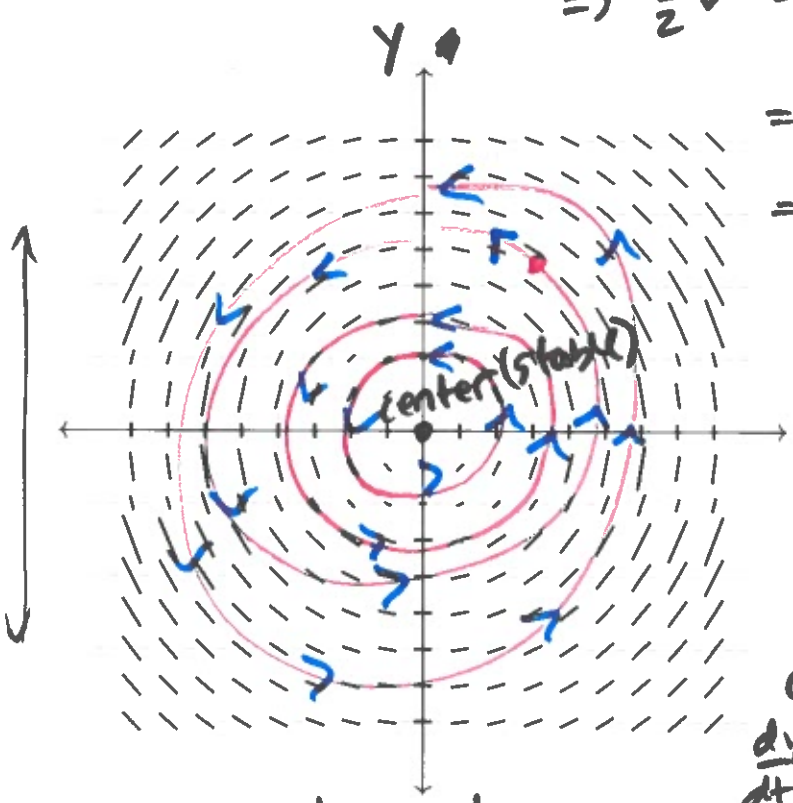
$\Rightarrow v^2 = -\frac{k}{m}y^2 + C$

$\Rightarrow v^2 + \frac{k}{m}y^2 = C$

$\Rightarrow \frac{v^2}{k} + \frac{y^2}{m} = C$

$\Rightarrow \left(\frac{v}{\sqrt{k}}\right)^2 + \left(\frac{y}{\sqrt{m}}\right)^2 = C$

ellipses.

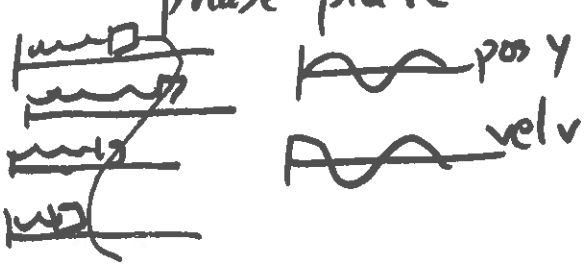


critical point?

$\frac{dy}{dt} = v = 0 \Rightarrow v = 0$ $(0,0)$

$\frac{dv}{dt} = -\frac{k}{m}y = 0 \Rightarrow y = 0$

phase plane



Example: If we look back at the homogeneous mass-spring oscillator with damping

$m, b, k > 0$

$$my'' + by' + ky = 0$$

we found the system

$r = \alpha + i\beta$

$\alpha < 0$

$\beta \geq 0$

$$\begin{cases} \dfrac{dy}{dt} = v \\[2ex] \dfrac{dv}{dt} = -\dfrac{k}{m}y - \dfrac{b}{m}v \end{cases}$$

$y_h = e^{\alpha t}\left(A\cos(\beta t) + B\sin(\beta t)\right)$

decay

periodic around crit pt

phase-plane equation: $\dfrac{dv}{dy} = \dfrac{-\frac{k}{m}y - \frac{b}{m}v}{v}$ homogeneous

y decaying sin

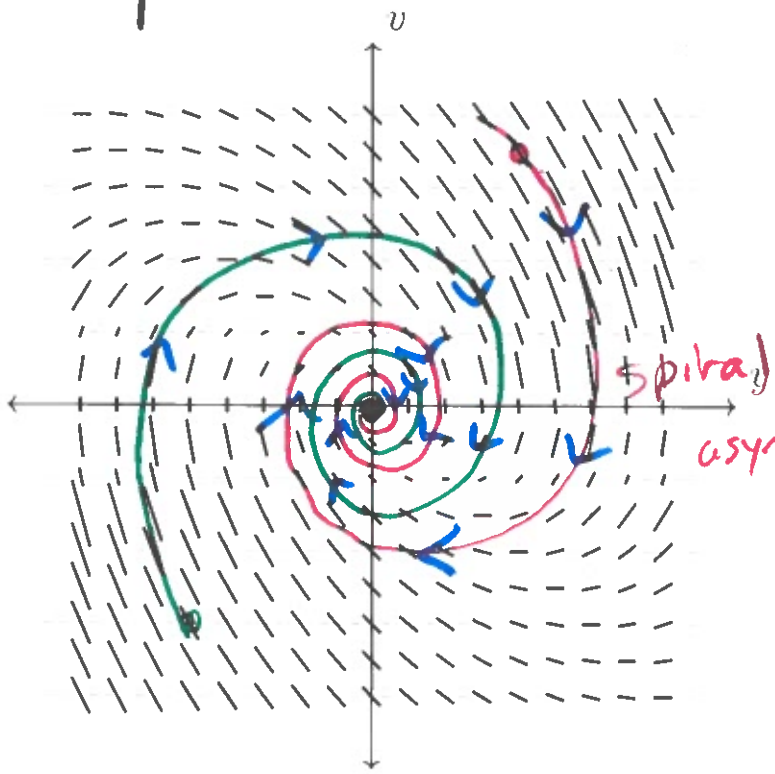
v decaying cosin

critical point:

$v = 0 \Rightarrow v = 0$

$-\frac{k}{m}y - \frac{b}{m}v = 0 \Rightarrow y = 0$

$(0,0)$



asymptotically stable